#12 · 3.4% of marks

Ch 3 — Advanced Capital Budgeting Decisions

SM TYK theory Qs: **certainty equivalent approach · sensitivity analysis · project appraisal under inflation · steps in simulation analysis.** Numericals: inflation NPV, ENPV / σ / CV, RADR, CE, sensitivity, scenario, decision tree, replacement cycle (EAC), APV.

① Current trends — factors (SM 1)

*Intermediate assumed certain cash flows; in practice projects face (i) **inflation** (ii) **change in technology** (iii) **change in government policies**.*

② Inflation in capital budgeting (SM 1.1) — TYK Q3

→ *Inflation raises revenues and costs; net revenues after adjusting for inflation = net revenues in current terms. Two distortions:*

→ ***(1) Depreciation on historical cost**: after-tax CF = (R − C)(1 − T) + DT — (R − C)(1 − T) moves with inflation, **DT does not** → inflation **reduces the real rate of return**. SM eg: ₹80,000, 4 yrs, revenue 60,000, costs 20,000, tax 50% → CF 30,000 p.a.; with 10% inflation nominal CF 32,000 → 39,282 but **real** CF 29,091 → 26,830 (falls)*

→ ***(2) Cost of capital contains an inflation premium**: $R_N = R_R + P$ (SM's formula). If the rate has an inflation premium, reflect inflation in cash flows; no inflation → discount at the real rate*

***Illus 1**: outlay 40,000, real CF 15,000 × 4 yrs, cost of capital 12% incl. 10% inflation premium. Discounting real CF at 12% is **inconsistent**. Either (i) nominal CF = 15,000 × 1.10^t at 12% → NPV **₹17,369**, or (ii) real rate = 1.12 ÷ 1.10 − 1 = **1.8%** on real CF → **₹17,367** — same answer.*

***Different inflation for revenue and cost**: $NPV = \Sigma\ [\{R_t\ \Pi(1 + i_r) - C_t\ \Pi(1 + i_c)\}(1 - T) + D_tT] \div (1 + k)^t - I_0$. **Illus 2**: ₹8 lakh, revenue inflation 10/9/8/7%, cost 12/10/9/8% (compounded), tax 60%, dep 2 lakh (shield 1.2 lakh, not inflated), k 10% → NPV **₹1,06,744**.*

⚠ TRAP

*Inflate cumulatively: year-3 revenue = R × 1.10 × 1.09 × 1.08. The depreciation tax shield is **never** inflated.*

③ Technology change (SM 1.2)

***Why analyse**: alters production process · benefits (quality, delivery, flexibility) · may lower cost of capital · better cash inflows · may need extra capex · shorter product life cycle. **How**: (1) at decision time use sensitivity / scenario / simulation (2) after launch, revise estimates (3) if unviable, evaluate **abandonment** (4) adjust the discount rate. Review continuously.*

4 Government policies (SM 1.3)

- → Affect the firm directly or via suppliers, buyers, service providers; positive or negative; revise estimates, consider abandonment. Some change rarely (New Industrial Policy 1991), others within a year: **fiscal policy** (spending, taxation, borrowing) and **monetary policy** (OMO, reserve requirements, selective credit control, currency supply, repo; MPC meets at least **4 times a year**)
- → **Domestic**: (a) interest-rate changes affect cost of debt / YTM — little effect on long-term-funded capex but big on **working capital** (overdraft); small changes ignored, ~**100 bps** calls for revision (b) fiscal policy changes tax and depreciation rates → revise cash flows
- → **International**: (a) bank rate and money supply move exchange rates (b) tax on foreign income / **DTAA** changes

5 Risk in investment decisions (SM 2)

- → Decisions under (i) **certainty** (ii) **risk** — probabilities can be assigned (iii) **uncertainty** — they can't
- → **Risk** = variability of actual vs estimated returns; measured by σ and CV. Risk: distribution known; uncertainty: not known (terms used interchangeably)
- → **Why adjust for risk**: (1) opportunity cost — are returns proportionate to risk vs other options? (2) to know the real value of inflows — higher risk, higher premium and return

6 Internal & external risk factors (SM 3)

Factor	SM meaning & example
Project-specific (internal)	Timely completion, estimation errors in resources & cash flows — nuclear vs hydel project
Company-specific (internal)	Rating downgrade, KMP changes, IPR / legal violations, labour disputes — two banks with different default exposure
Industry-specific	Regulatory restrictions, technology — leather, breweries
Market	Substitutes, demand, access to resources — coal supply to a thermal plant
Competition	Rival entry, product dynamism, change in tastes
Economic conditions	Monetary & fiscal changes (new taxes, cess), inflation, GDP, savings, disposable income
International	Trade / market-access restrictions, recession, bilateral agreements, political & geographical — limits on outsourcing jobs abroad

7 Techniques of risk analysis (SM 4)

Statistical

Probability · variance / σ · coefficient of variation

Conventional

Risk-adjusted discount rate · certainty equivalent

Others

Sensitivity · scenario · simulation · decision tree

TYK THEORY Q1

⑧ Statistical techniques (SM 4.1)

ENCF = $\Sigma\ NCF_i \times P_i$ · **ENPV** = $\Sigma\ ENCF_t \div (1 + k)^t$ (incl. outflows) · $\sigma = \sqrt{\Sigma\ P_i(NCF_i - ENCF)^2}$ · **CV = σ ÷ expected return / cash flow**

CV = risk per unit of expected return — compare projects of different size; **lower CV preferred**. SM eg: 3,00,000 × 0.3 + 2,00,000 × 0.6 + 1,20,000 × 0.1 = **₹2,22,000**.

⑨ Risk-adjusted discount rate (SM 4.2.1)

→ Investors demand more from risky projects: rate = compensation for delaying consumption + inflation (= **risk-free**) + **risk premium** (depends on perceived risk & risk aversion). **RADR = R_f + risk premium**

→ **Illus 7**: ₹100 lakh, CF 25/60/75/80/65 → at 7% NPV **144.34**; at RADR 14% NPV **99.79**

→ **Advantages**: (1) easy to understand (2) builds the premium into the discount rate. **Limitations**: (1) hard to find the premium / RADR (2) NPV yes, but **σ of the project can't be computed**

⑩ Certainty equivalent (SM 4.2.2) — TYK Q1

CIMA: expressing risky future cash flows as the certain cash flow the decision-maker would consider equivalent — **indifferent** between the risky amount and the (lower) riskless amount. Lets the decision-maker build in his **utility function**; a set of riskless cash flows replaces the original.

1. **Step 1**: α_t = certain CF ÷ risky (expected) CF; multiply each CF by α_t. Coin toss: win ₹10,000 or nothing; indifferent at ₹3,000 certain → **CE ₹3,000, α = 0.3**
2. **Step 2**: discount at the **risk-free rate** — cost of capital would **double-count** risk
3. **Step 3**: normal capital budgeting; for **IRR compare with risk-free rate**, not the required return

α between 0 and 1; α = 1 → certain or risk-neutral; industry usually risk-averse. **NPV = $\Sigma\ \alpha_t\ NCF_t \div (1 + k_f)^t - 1$**. **Illus 8**: ₹45 lakh; 10/15/20/25 lakh × 0.90/0.85/0.82/0.78 at 5% → **₹5,34,570**.

Advantages

(1) simple to understand and apply (2) easily set for different risk levels in different years

Disadvantages

(1) no objective / mathematical way to estimate — subjective (2) reflects **management's** risk perception, ignores **shareholders'** → rarely used

RADR vs CE: CE is **superior** — it doesn't assume risk grows with time at a constant rate (each year's α fits its risk). But RADR is preferred in practice — easier to adjust a rate than specify a series of α.

11 Sensitivity analysis (SM 4.3.1) — TYK Q2

- → **CIMA**: modelling and risk-assessment procedure in which changes are made to significant variables to find their effect on the planned outcome; special attention to variables of special significance
- → Change **one variable at a time**; the more sensitive the NPV (IRR), the more **critical** the variable. **Steps**: (1) find variables influencing NPV / IRR (2) establish mathematical relationships (3) analyse effect of change in each
- → SM Illus 9: 2.5% adverse change in each variable → **selling price** has the maximum effect (NPV −24.82%)
- → **Advantages**: (1) identifies **critical issues** (2) **simple**. **Disadvantages**: (1) assumes variables are **independent** (2) **ignores probability** of changes

12 Scenario analysis (SM 4.3.2)

Extends sensitivity: brings in **probabilities** and changes **several variables at once**. Base (most likely) case → worst case (low sales & price, high variable cost) → best case; or mixed. Variability of outcomes across scenarios (eg boom / normal / recession) measures risk. **More complex** than sensitivity — all inputs change simultaneously.

13 Simulation — Monte Carlo (SM 4.3.3) — TYK Q4

Exact replica of the actual situation via a model; infinite calculations for possible outcomes and probabilities; ties together sensitivities and probability distributions; named after casino gambling (from first nuclear-bomb work); gives a **probability distribution of NPV**, not a point estimate.

1. **Model the project** — NPV vs **parameters** (set by decision-maker, constant across runs) and **exogenous variables** (stochastic, outside control)
2. Specify parameter values and probability distributions of exogenous variables
3. Select a value **at random** from each exogenous distribution
4. Compute NPV for those values
5. **Repeat** 3 & 4 many times
6. Plot the NPV distribution; compute **mean and σ**

Advantages — "strength lies in variability"

(1) predict bad market situations beforehand (2) handles many exogenous variables with any distribution, complex inter-relationships beyond analytical methods, forces explicit consideration of inter-dependencies

Shortcomings (4)

(1) hard to model and specify distributions (2) inherently **imprecise** — misleading when a tail matters (3) built by experts, decision-maker may not understand / use it (4) NPV at **risk-free** rate (to avoid pre-judging risk) → hard to interpret

14 Decision tree (SM 4.3.4)

- → Graphic display of the link between a present decision and future events, decisions and consequences — for **sequential** decisions. **Decision** (manager controls; acts rationally) vs **event** (no control; assign probabilities)
- → **Steps**: (1) define investment (launch local / national / international; extend factory vs outsource; dig for oil) (2) identify alternatives (3) draw the tree (4) evaluate — from the **last decision (right) backwards**; at each decision point keep highest NPV, truncate others; compute expected NPV at chance points
- → Rules: decision node = **rectangle**; chance node = **circle**; alternatives as straight lines; drawn **left to right**; number nodes; add values & probabilities

15 Replacement decisions (SM 5)

→ **Replace existing machine** (newer version or a different machine with the same function), mostly to maintain existing levels of operation. Method (app-written): incremental NPV — cost of new less sale of old, change in operating cash flows and depreciation tax shield, terminal values

→ **Optimum replacement cycle** (continuous replacement): NPVs of different cycle lengths aren't comparable → **EAC = PV of costs ÷ PVAF for the cycle**; choose the **lowest EAC**. SM taxi: EAC 1 yr 3,40,044 · 2 yr **3,15,331** · 3 yr 3,17,639 → replace every **2 years**. ICAI MTP machine: 4-year cycle lowest at ₹58,656

16 Adjusted Present Value (SM 6)

APV = Base-case NPV + PV of tax benefit on interest. Base case at the **unlevered** cost of equity (all-equity firm; unlevered K_e from Ch 13). **Separates investment and financing decisions**; credits the project for the debt used to finance it.

17 Correlation — how Ch 3 actually connects

Ch 3 idea	The actual link (app-written cross-reference; the Ch 3 side is SM)
Real vs nominal (1.12 ÷ 1.10 − 1)	**Ch 5**: equity is valued on **nominal** CF at nominal rates; **Ch 11**: PPP uses the same inflation ratio to forecast exchange rates.
RADR = R_f + premium	**Ch 6**: CAPM gives the premium objectively — $\beta(R_m - R_f)$; **Ch 11**: SM says don't load political / FX risk into the rate — use certainty equivalents.
Certainty equivalent at R_f	**Ch 9**: risk-neutral valuation also discounts at R_f after adjusting the cash flows / probabilities — same "adjust numerator, not denominator" logic.
σ and CV of NPV	**Ch 6**: σ and CV of returns; **Ch 2**: σ × 2 = VaR.
Abandonment, decision tree	**Ch 9**: real options (abandon, expand, wait) give value that static NPV misses; decision trees value them.
APV	**Ch 11**: SM's APV for foreign projects adds interest subsidies; **Ch 13**: unlevered K_e via asset beta.
Replacement cycle / EAC	**FM**: EAC from Intermediate; equal-life comparison logic as in lease vs buy.